Joseph Walton

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Professional Summary

Senior AI and machine learning engineer with over 10 years of experience delivering production systems across regulated government, financial services, intellectual property and defence environments. Hands-on expertise in Agentic systems, RAG, LangChain, LangGraph and Azure AI services. Proven technical leader across architecture, implementation, evaluation, observability and deployment, with delivered solutions saving approximately millions annually. Strong advocate for secure, responsible AI who communicates effectively with engineering, architecture, business and international stakeholders.

Technical Skills

- **Generative AI:** Azure OpenAI, Azure AI Foundry, LangChain, LangGraph, RAG, agentic AI, LLM evaluation and observability
- **Programming:** Python, SQL, Java
- **Search & Data:** Elasticsearch, vector search, HNSW, Apache Spark
- **MLOps & CI/CD:** Azure ML Pipelines, Vertex AI Pipelines, MLFlow, Docker, MLOps
- **Cloud Platforms:** Microsoft Azure (Functions, AI Search, Cognitive Services, Azure Databricks)
- **Machine Learning:** HuggingFace, TensorFlow, PyTorch, Scikit-learn
- **Security:** AI red teaming, adversarial testing, secure containerised deployment, responsible AI

Professional Experience

IBM

Lead Data Scientist
June 2025 - Present | Contract role delivering AI solutions for central government

- Designed and delivered a production agentic data-processing solution with Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Rescued underperforming AI projects by introducing systematic evaluation and observability practices.
- Built a Claude Skill to streamline human-in-the-loop data ingestion for LLM evaluation, saving approximately one hour per evaluation case.
- Re-engineered a data-matching algorithm in Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, Azure AI Foundry, Azure Functions, LangGraph, Polars

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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

- Developed production Generative AI applications using LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent to help users identify relevant Goods and Services terms.
- Implemented and tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation.
- Improved search and recommendation relevance metrics by 20% through data-driven experimentation and optimisation.
- Implemented an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Conducted red-team exercises to assess Generative AI applications against adversarial attacks and guide robustness improvements.
- Advised technical and non-technical stakeholders across international intellectual property offices and contributed to cross-government AI policy and best practices.
- Replaced Azure Databricks ETL workloads with Polars pipelines, cutting compute costs by 50% and enabling zero-downtime deployments.

Tech Stack: Python, Azure OpenAI, LangChain, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

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Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

- Led cross-functional delivery of secure, production machine learning solutions using large-scale text and image data.
- Designed and implemented an MLOps framework on Google Cloud Platform, orchestrating training, evaluation, approval and deployment with Vertex AI Pipelines and reducing time to market to under one day.
- Developed and deployed a multilingual BERT model that detected intellectual property infringements across more than 90 languages.
- Built Spark pipelines for model training, batch inference and recommendation workloads.
- Mentored a software engineer through a transition into a machine learning role.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Engineered a production NLP system in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image-similarity model that helped secure a major blue-chip contract representing 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

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Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed Python machine learning services and scalable data pipelines to integrate and process sensitive administrative datasets from multiple sources.
- Worked in a cross-functional team to develop a highly scalable backend for the UK Census 2021 data-collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018 | Secondment

- Applied statistical and analytical methods to securely and ethically use payments data for regional economic indicators.
- Continued work from the winning ONS x Barclays Hackathon team.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed Python and Java big-data processing systems for government and defence clients.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner, ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).